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Desirable Price Function and Value Ratio of Constant Function Market Makers

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***Abstract*—Decentralized exchange was a revolution in cryptocurrency trading. Decentralized exchange presented a method to set prices mathematically without an order book. Order books have been a widely-known method of determining prices for over 400 years. Decentralized exchanges devised a constant function market maker to mathematically determine prices. Among them, constant product market makers are the most widely used. In addition, several constant function market makers have been proposed. However, there was no discussion about the desirable requirements of a constant-function market maker. In this paper, we discuss the desirable requirements of a constant function market maker and show that the constant product market maker is still the most desirable constant function market maker.**

Keywords—blockchain, cryptocurrency, decentralized exchange, market maker, smart contract

I. Introduction

Order book-based trading systems have a history of more than 400 years. Blockchain-based decentralized exchanges have shown that exchanges without order books are possible with three innovations: automated market maker, smart contract, and liquidity pool. The automated market maker is a market maker operating automatically without human work. In that sense, the market makers are said to be automated. On the other hand, the automated market makers let customers remove the price from their ask or bid orders and only quote the quantity. The price is determined automatically and mathematically by the automated market makers. The automated market makers are implemented using smart contracts. To ensure sufficient liquidity to enable trading whenever desired, automated market makers introduced liquidity pools.

Bancor [1] invented an automated market maker liquidity pool on the Ethereum network in 2017. Bancor founders played an instrumental role in creating the Bancor decentralized exchange, a network of on-chain liquidity pools to pair pools of assets with one another. The Bancor protocol was named after Keynes' Bretton Woods international reserve currency proposal in 1944. Users can trade against these Bancor pooled asset pairs, with prices set mathematically. The liquidity pool smart contract makes it possible for traders to buy and sell assets at any time and immediately since each liquidity pool has a pair of base and pricing assets deposited. For example, Ethereum (ETH) is the base asset to be traded, and Tether (USDT) is the pricing asset (a.k.a. stablecoin) in the liquidity pool. The price of the trading asset is set mathematically by the smart contract, irrespective of the intention of the buyer or seller. Thus, asset trading can be completed in an instant due to the abundant liquidity and mathematical pricing.

A few years before Bancor, Hanson [2],[3] introduced LMSR (logarithmic market scoring rule) for the prediction market. The LMSR can be used as an automated market maker [4]. Othman *et al.* [5] proposed LS-LMSR (Liquidity-sensitive logarithmic market scoring rule), which is an advanced version of LMSR. However, the LMSR-type automatic market makers require calculations that are too computation-intensive to determine prices with smart contracts, but they also have several problems.

Uniswap [6] adopted CPMM (constant product market maker) is sufficiently popular, effective and sustainable. Balancer [9] introduced the CMMM (constant mean market maker) that generalized the CPMM and has reduced impermanent loss. In Wang's study [4], CSMM (constant sum market maker) and CCMM (constant circle market maker) are covered. Angeris and Chitra [8] introduced a constant function market maker model.

Impermanent loss is like a penalty imposed on liquidity providers. A phenomenon in which liquidity providers benefit more from not depositing assets than depositing them is called impermanent loss. Most liquidity providers provided liquidity despite impermanent loss due to interest and governance token rewards for them. Thus, all AMMs have tried and failed to remove impermanent loss. Recently, Kim *et al.* [9][9] computed the impermanent loss of four automated market makers under the assumption that price is the ratio of the number of two assets and showed that impermanent gain is achievable. However, this assumption is effective only for CPMM market makers.

In this paper, we introduce two important concepts: the price function and the value ratio to determine which one is desirable as an automated market maker. Based on these concepts, we can see which automated market makers are useful. We can conclude that some automated makers cannot be used.

II. Mathematical Price

In this paper, all automated market maker functions consider only a constant cost function containing two variables x and y. The constant cost function such as $C(\mathbf{q}) = k$ is called constant function market maker, where k is a constant. A state $\mathbf{q}$ has two variables x and y, representing the number of two outstanding tokens. Hanson's LMSR automated market maker [2][3] was originally proposed for the prediction markets. Thus, his price concept meets the requirement for prediction markets. Since his prices represent the probability of winning in the prediction, the sum of all prices had to be 1. Othman *et al.* defined this property as translation invariance [5].

The price p_x of x and the price p_y of y are obtained by differentiating the cost function as follows.

$$p_x = \frac{dC(\mathbf{q})}{dx}, \quad p_y = \frac{dC(\mathbf{q})}{dy}.$$

When the sum of the two prices is 1 (i.e. $p_x + p_y = 1$), the price is defined as translation invariant. Therefore, since the price of the stablecoin is already set at 1, it is unreasonable to use this price on the exchange.

Decentralized exchange automatic market makers use a new pricing system. The price of the stablecoin is set at 1, so P_y is 1. P_x is determined as the ratio of the difference between x_{t-1} and x_t and the difference between y_{t-1} and y_t. In other words,

$$P_x(\mathbf{q_t} - \mathbf{q_{t-1}}) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}}.$$

If the difference between x_{t-1} and x_t is infinitesimal, we have

$$P_x(\mathbf{q_t}) = \lim_{x_t \to x_{t-1}} -\frac{y_t - y_{t-1}}{x_t - x_{t-1}} = -\frac{dy_t}{dx_t} = \frac{p_x}{p_y}.$$

In this paper, unit price, P_u, is defined. Since y coins are stablecoins, the price of all y coins is y dollars. Under the assumption that the value of x coins and y coins must be equal, the unit price is defined as follows:

$$P_u(\mathbf{q_t}) = \frac{y_t}{x_t}.$$

The value of the assets in the automated market makers is given as a product of asset price and number as follows:

$$V_x = P_x \cdot x, \quad V_y = y.$$

One of the important measures is the value ratio, V_r. It is desirable for this ratio to be constant regardless of the state $\mathbf{q}$. The value ratio is defined as follows:

$$V_r = \frac{V_x}{V_y} = \frac{P_x \cdot x}{y}$$

and

$$V_r = \frac{V_x}{V_y} = P_x \cdot P_u.$$

III. LMSR and LS_LMSR

Hanson proposed the LMSR market maker [2][3] which has the cost function as follows:

$$C(\mathbf{q}) = b \cdot \ln\left(e^{\frac{x}{b}} + e^{\frac{y}{b}}\right) = k$$

for a constant b and $\mathbf{q} = (x, y)$, where x and y represent the number of two outstanding tokens.

Its price function for each token is given as follows:

$$p_x(\mathbf{q}) = \frac{dC(\mathbf{q})}{dx} = \frac{e^{\frac{x}{b}}}{\left(e^{\frac{x}{b}} + e^{\frac{y}{b}}\right)}$$

and

$$p_y(\mathbf{q}) = \frac{dC(\mathbf{q})}{dy} = \frac{e^{\frac{y}{b}}}{\left(e^{\frac{x}{b}} + e^{\frac{y}{b}}\right)}$$

and

$$P_x(\mathbf{q}) = \frac{p_x}{p_y} = \frac{e^{\frac{x}{b}}}{e^{\frac{y}{b}}} = \frac{e^{\frac{x}{b}}}{e^{\frac{k}{b}} - e^{\frac{x}{b}}}.$$

Here, it is clear that the price range is limited to between 0 and 1, such that

$$0 < p_x, p_y < 1.$$

It is not realistic that the price upper limit is set below 1. The parameter b in the LMSR cost function determines the degree of liquidity sensitivity. As the value of b increases, the market makers become less sensitive and the price changes more slowly. It is not easy to properly adjust the value of b. Once the value of b is set, there is no change in sensitivity, so LMSR is said to be insensitive.

The value of each asset in the LMSR automated market maker is given as follows:

$$V_x = \frac{e^{\frac{x}{b}}}{e^{\frac{k}{b}} - e^{\frac{x}{b}}} \cdot x,$$

and

$$V_y = y = b \ln\left(e^{\frac{k}{b}} - e^{\frac{x}{b}}\right)$$

and

$$V_r = \frac{e^{\frac{x}{b}}}{e^{\frac{k}{b}} - e^{\frac{x}{b}}} \cdot \frac{x}{b \ln\left(e^{\frac{k}{b}} - e^{\frac{x}{b}}\right)}.$$

Then, it is trivial to see $V_r(\mathbf{q}_i) \neq V_r(\mathbf{q}_j)$ for different i and j.

Othman *et al*. [5] pointed out that Hanson's LMSR price is not liquidity sensitive, and he proposed an LS-LMSR market maker that is liquidity sensitive. The LS-LMSR cost function is given as follows:

$$C(\mathbf{q}) = \alpha(x+y) \cdot \ln\left(e^{\frac{x}{\alpha(x+y)}} + e^{\frac{y}{\alpha(x+y)}}\right).$$

Notice that b in the LSMR is replaced with $\alpha(x+y)$ in the LS-LMSR. LS-LMSR is said to be sensitive because the sensitivity changes whenever the value of $\alpha(x+y)$ changes.

In this case,

$$p_x = \alpha\ln\left(e^{\frac{x}{\alpha(x+y)}} + e^{\frac{y}{\alpha(x+y)}}\right) + \frac{y\left(e^{\frac{x}{\alpha(x+y)}} - e^{\frac{y}{\alpha(x+y)}}\right)}{(x+y)\left(e^{\frac{x}{\alpha(x+y)}} + e^{\frac{y}{\alpha(x+y)}}\right)}$$

and

$$p_y = \alpha\ln\left(e^{\frac{x}{\alpha(x+y)}} + e^{\frac{y}{\alpha(x+y)}}\right) + \frac{x\left(e^{\frac{y}{\alpha(x+y)}} - e^{\frac{x}{\alpha(x+y)}}\right)}{(x+y)\left(e^{\frac{x}{\alpha(x+y)}} + e^{\frac{y}{\alpha(x+y)}}\right)}$$

Thus, as is derived in [4],

$$P_x(\mathbf{q}) = \frac{p_x}{p_y} = \frac{\alpha(x+y)(X+Y)\ln(X+Y) + y(X-Y)}{\alpha(x+y)(X+Y)\ln(X+Y) + x(Y-X)}$$

where

$$X \equiv e^{\frac{x}{\alpha(x+y)}}, \qquad Y \equiv e^{\frac{y}{\alpha(x+y)}}$$

and

$$P_y(\mathbf{q}) = 1.$$

As a constant-function market maker, LS-LMSR has one serious problem. First, given the value of x, it is not easy to find the value of y in the following equation.

$$C(\mathbf{q}) = \alpha(x+y) \cdot \ln\left(e^{\frac{x}{\alpha(x+y)}} + e^{\frac{y}{\alpha(x+y)}}\right) = k.$$

Although it is possible to obtain it through numerical analysis, it consumes an excessive amount of gas when calculations are performed in smart contracts.

The above equation for $\alpha = 1$ can be expressed as follows:

$$e^{\frac{x}{x+y}} + e^{\frac{y}{x+y}} = e^{\frac{k}{x+y}}.$$

This exponential equation can be reformulated as a polynomial form as follows:

$$z^2 + e = z^a,$$

where

$$s \equiv \frac{y}{x}, \qquad a \equiv \frac{k}{x}, \qquad z \equiv e^{\frac{1}{1+s}}.$$

Second, price P_x does not function as a valid price because it violates the principle of supply and demand. In other words, as the value of x decreases (i.e., as it becomes rare), P_x should increase, and as the value of x increases (i.e., as it becomes abundant), P_x should decrease. The reason why price P_x does not satisfy the law of supply and demand is because of the concavity of cost function $C(\mathbf{q})$.

For example, when x is 10, y is 10 and $\alpha = 1$, the value of k is 23.862943. In this case, P_x is 1. When the value of k remains at 23.862943, if x is set to 1, y is 17.506184. In this case, note that P_x is 0.680982. In the previous case, there were 10 units of asset and its unit price was 1. Now, there is 1 unit of asset and its unit price is 0.680982, which is lower than 1. The market principle is that the rarer something is, the higher the price should be.

Third, when α is 1, the range of price P_x is narrow, ranging from 0.6481 to 1.5430. Moreover, no matter what value k is, as long as it is a positive number, the range is the same. The price cannot be lowered below the lower limit and cannot be raised above the upper limit. By adjusting the value α, we can control the price range.

LMSR and LS-LMSR market makers are good for trading a pair of stablecoin and a fiat currency. In normal times, the price volatility of stablecoins should not be large. For the LMSR, the price range is obtained as follows:

$$\frac{1}{e^{\ln(e^{k/b}-1)}} \leq P_x(\mathbf{q}) \leq e^{\ln(e^{k/b}-1)}.$$

Thus, as the value of b increases, the volatility of price P_x decreases. The range depends on two parameters: b and k.

In the case of LS-LMSR, the range of price can be controlled by adjusting the parameter α. The price range is computed as follows:

$$\frac{\alpha \cdot \ln\left(1+e^{\frac{1}{\alpha}}\right)\left(1+e^{\frac{1}{\alpha}}\right) + \left(1-e^{\frac{1}{\alpha}}\right)}{\alpha \cdot \ln\left(1+e^{\frac{1}{\alpha}}\right)\left(1+e^{\frac{1}{\alpha}}\right)} \leq P_x(\mathbf{q}) \leq \frac{\alpha \cdot \ln\left(1+e^{\frac{1}{\alpha}}\right)\left(1+e^{\frac{1}{\alpha}}\right)}{\alpha \cdot \ln\left(1+e^{\frac{1}{\alpha}}\right)\left(1+e^{\frac{1}{\alpha}}\right) + \left(1-e^{\frac{1}{\alpha}}\right)}.$$

IV. CPMM

CPMM cost function, which is used in Uniswap [6], is a product of x and y as follows:

$$C(\mathbf{q}) = x \cdot y = k.$$

Then,

$$p_x = \frac{dC(\mathbf{q})}{dx} = y, \qquad p_y = \frac{dC(\mathbf{q})}{dy} = x.$$

Thus,

$$P_x(\mathbf{q}) = \frac{p_x}{p_y} = \frac{y}{x} = \frac{k}{x^2}.$$

It is clear that price P_x satisfies the law of supply and demand. That is, as the value of x decreases (quantity decreases), the price increases, and vice versa. It is because of the convexity of cost function $C(\mathbf{q})$. The price P_x monotonically decreases as x increases. This observation is natural. For the CPMM, note that $V_r(\mathbf{q}_i) = V_r(\mathbf{q}_j) = 1$ for any i and j.

Uniswap V3 uses constrained liquidity, such as

$$\left(x + \frac{k}{\sqrt{P_b}}\right)\left(y + k\sqrt{P_a}\right) = k^2.$$

This constant product market maker is effective for

$$0 \le x \le k\frac{\sqrt{P_b} - \sqrt{P_a}}{\sqrt{P_a} \cdot \sqrt{P_b}}$$

and its price range is limited to

$$P_a \le P_x(\mathbf{q}) \le P_b.$$

The price of the Uniswap V3 market maker is represented as follows:

$$P_x(\mathbf{q}) = \frac{p_x}{p_y} = \frac{y + k\sqrt{P_a}}{x + \frac{k}{\sqrt{P_b}}} = \frac{k^2}{\left(x + \frac{k}{\sqrt{P_b}}\right)^2}$$

The value function of Uniswap V3 is computed as follows:

$$V_r = \frac{k^2 \cdot x}{\left(x + \frac{k}{\sqrt{P_b}}\right)\left(k^2 - k\sqrt{P_a}\left(x + \frac{k}{\sqrt{P_b}}\right)\right)}.$$

Note that the value function of Uniswap V3 is not a constant.

V. CSMM

CSMM cost function cannot be used as an automated market maker because its price function is constant no matter what the state is. Its cost function is given as follows:

$$C(\mathbf{q}) = x + y = k.$$

Its price function is given as follows:

$$p_x = 1, \qquad p_x = 1, \qquad P_x = 1.$$

Its value ratio is not constant. Its value ratio is expressed as follows:

$$V_r(\mathbf{q}) = \frac{V_x}{V_y} = \frac{x}{y} = \frac{x}{x - k}.$$

Since its value ratio is not constant, it is not easy to use the CSMM automated market maker for liquidity management. The CSMM cost function is neither convex nor concave. Thus, CSMM does not incur an impermanent loss. No impermanent loss is a desirable property of an automated market maker. CSMM is the only one that does not cause impermanent loss among many automated market makers.

VI. CMMM

CMMM is currently being used in a decentralized exchange, Balancer [7]. CMMM's cost function has one desirable property. CMMM is better than CPMM in terms of impermanent loss. CMMM has a milder degree of impermanent loss than CPMM. CMMM cost function is given as follows:

$$C(\mathbf{q}) = x^\alpha \cdot y = k.$$

Thus, CPMM is a special case of CMMM with $\alpha = 1$.

For CMMM, the price function is given as follows:

$$p_x = \alpha x^{\alpha-1} \cdot y, \qquad p_y = x^\alpha,$$

and

$$P_x = \frac{\alpha x^{\alpha-1} \cdot y}{x^\alpha} = \alpha\frac{k}{x^{\alpha+1}}.$$

Thus,

$$V_x = \alpha\frac{k}{x^{\alpha,}}. \; V_y = y$$

and

$$V_r = \alpha\frac{k}{x^\alpha \cdot y} = \alpha.$$

It shows that CMMM also has a constant value ratio function as CPMM, while (LS-)LMSR and CSMM market makers have a non-constant value ratio.

VII. CCMM

The CMMM cost function is given as follows: as follows:

$$C(\mathbf{q}) = x^2 + y^2 = k^2.$$

This cost function is effective in the non-negative range as follows:

$$0 \le x, y \le k.$$

In this case, the CCMM automated market maker is concave. In this case, the price function is derived as follows:

$$p_x = 2x, \qquad p_y = 2y,$$

and

$$P_x = \frac{x}{y} = \frac{x}{\sqrt{k^2 - x^2}}.$$

It means that the price function violates the demand and supply principle. The price function is monotonically increasing with respect to x. The value ratio of CCMM is given as follows:

$$V_r = \frac{x^2}{k^2 - x^2}.$$

The value ratio is monotonically increasing with respect to x.

VIII. Conclusion

In this paper, we examined well-known automated market makers, including LSMR, LS-LMSR, CPMM, CSMM, CMMM, and CCMM. Each market maker has its pros and cons. In order to analyze the pros and cons, we defined the price functions and value ratios of the market makers. Then, we calculate the price function and value ratio of each market maker. After that, we analyze their characteristics.

The cost functions of LMSR and LS-LMSR, CCMM are concave, and their price functions violate the demand and supply principle. Their price functions are monotonically increasing with respect to x. In order to make the price function satisfy the principle of demand and supply, as an alternative, we suggest the reciprocal of the monotonically increasing price as a desirable price. On the other hand, the cost function of CPMM and CMMM is convex and their price function is monotonically decreasing. Desirable cost functions have a concave curvature and their price functions satisfy the principle of demand and supply.

The value ratio of LMSR and LS-LMSR, CSMM, and CCMM is not constant. On the other hand, CPMM and CMMM's value ratio is a constant, such that $V_r = \alpha$. Note that when the value ratio is constant, it is easy and intuitive to calculate the number of assets to deposit or withdraw a pair of assets in the liquidity pool. The desirable value ratio is a constant.

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